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EXPERIMENT 8

PERFORM WEB SPIDERING

AIM:

To automatically map out the complete structure of a target website - every page, link, and form - using a web spidering (crawling) tool.

THEORY:

Web spidering (also called crawling) is the process of automatically following every link on a website to discover all its pages, directories, and forms — similar to how Google's crawler indexes the internet, but done locally against one target for security testing purposes.

Manually clicking through every page of a large application is slow and error-prone. A spider tool automates this by:

1. Starting from a given URL
2. Extracting all links/forms on that page
3. Visiting each new link found
4. Repeating until no new links are discovered

TOOLS REQUIRED

- Burp Suite (Spider/Crawler feature, in the Community or Pro edition)
- OWASP ZAP (alternative, has a free built-in spider)
- Target: DVWA or OWASP Juice Shop

PROCEDURE

Step 1: Set Up the Target in Burp Suite

1. Configure the browser proxy to route through Burp (127.0.0.1:8080) as done in Experiment 7.
2. Browse to the home page of the target application once, so it appears under Burp's **Target → Site map**.

Step 2: Add the Target to Scope

1. Right-click the target's entry in the Site map → **Add to scope**.
2. This tells Burp to only crawl/act on this specific application, not the whole internet.

Step 3: Run the Crawler

(In Burp Suite Community Edition, this may be a manual "Spider" via passive browsing; Burp Pro has an automated Crawl button.)

If using **OWASP ZAP** instead (fully free automated spider):

```
zaproxy
```

1. Enter the target URL in the **Quick Start** tab.
2. Click **Spider** to begin the automated crawl.
3. Watch the **Spider** tab — it will list every URL discovered in real time.

Step 4: Review the Discovered Site Map

Once the spider finishes, expand the **Sites** tree in ZAP (or Site map in Burp) to see the full structure: every folder, page, and parameter discovered.

Export the list of discovered URLs:

```
Right-click site root → Export → Export URLs to File
```

Step 5: Identify Forms Found During Spidering

Go through the discovered pages and note down every HTML form found (login, search, comment, upload), along with its input field names — this list becomes the target list for vulnerability scanning in Experiment 9.

Step 6: Record the Site Map Summary

Category	Count / Details
Total Pages Discovered	42
Forms Found	Login, Search, Comment
Parameters Found	id, user, comment
Hidden/Unlinked Pages Found	/admin/settings.php

OUTPUT

A complete, exported site map of the target application listing every discovered page, along with a list of forms and parameters found — ready to be used as input for automated vulnerability scanning.

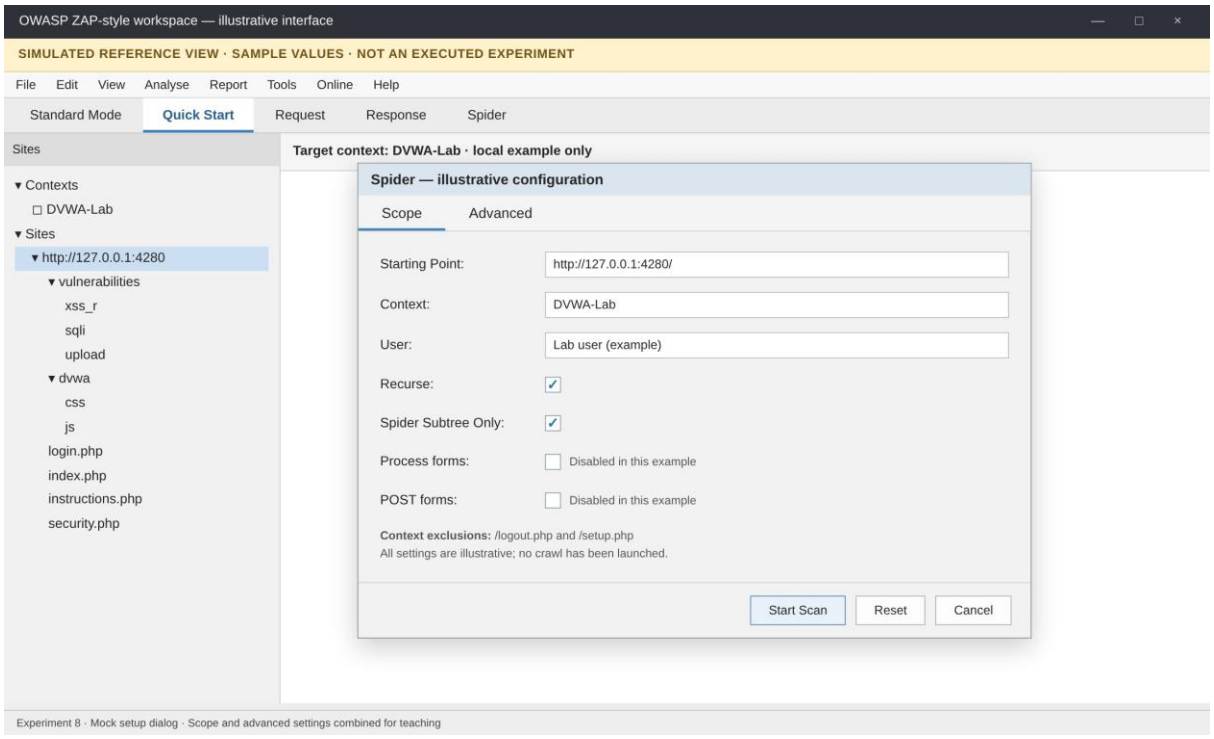


Fig. 1: Spider configuration (simulated reference).

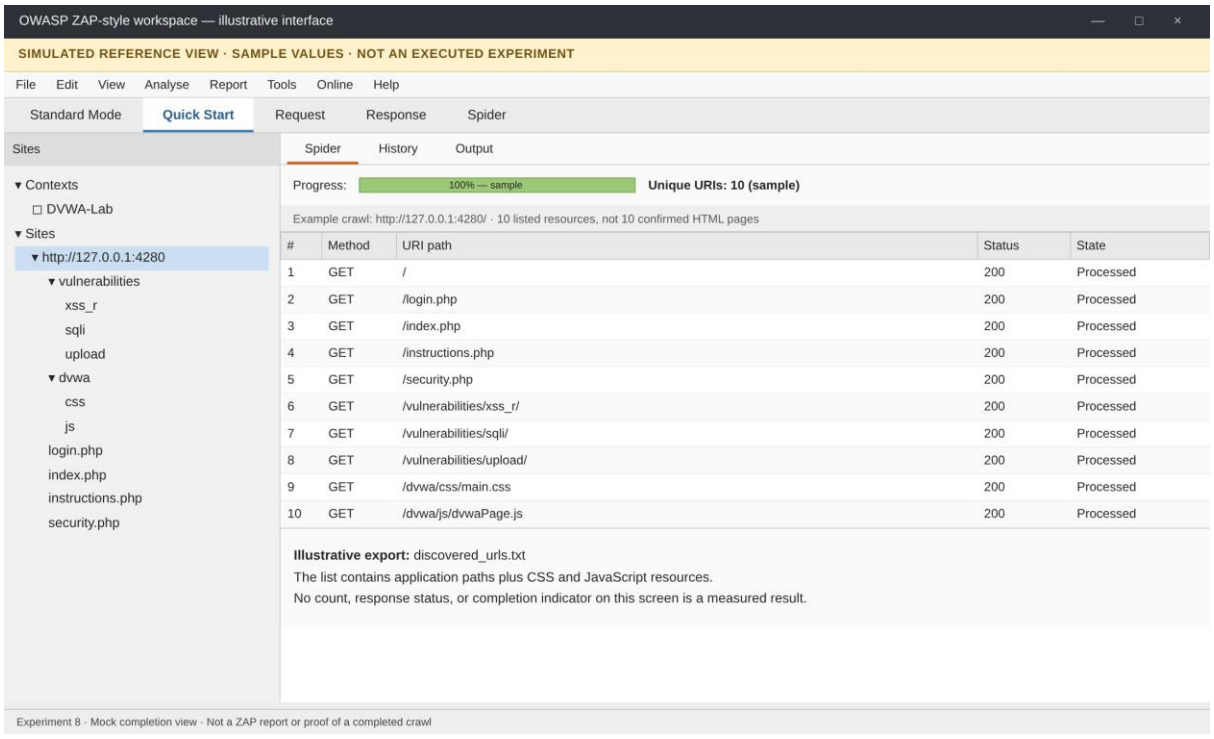


Fig. 2: Discovered site map (simulated reference).